



SYMBIOSIS CENTRE FOR INFORMATION TECHNOLOGY (SCIT), PUNE

(A constituent of the Symbiosis International University, Pune)

DELIVERY PERFORMANCE TRACKER

Carrier and region analytics for on-time delivery, delay, freight cost per kg and damage/RTO rate, with a weighted carrier scorecard and a savings scenario (Task T12 — Mini Project)

SEMESTER-I

AI Powered Tools

Submitted By:

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Date: 04-10-2026

Build Evidence

The tracker is a single self-contained HTML file (browser-side JavaScript, no server). The user uploads the Shipments sheet exported as CSV; no data is built into the file, and no numbers are shown until a file is loaded. Figures 1–4 are screenshots of the final application; Figures 5–8 (scorecard, savings, recommendation, email) are in Appendix A.

Build Link / HTML File: file:///Users/dheemhansaravanan/Downloads/delivery_performance_tracker.html

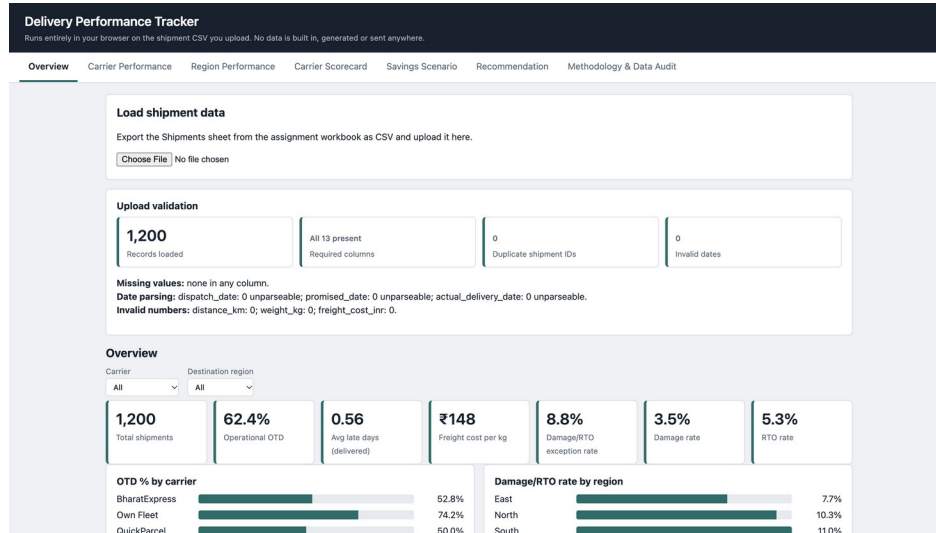


Figure 1 — Overview tab: upload validation (1,200 records, 13/13 columns, 0 duplicates, 0 invalid dates), seven KPI cards and carrier / region filters

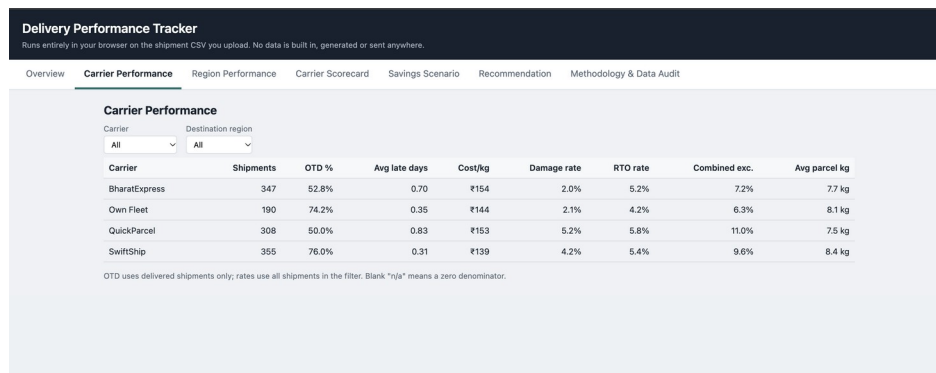


Figure 2 — Carrier Performance tab: carrier KPI table with carrier and destination-region filters

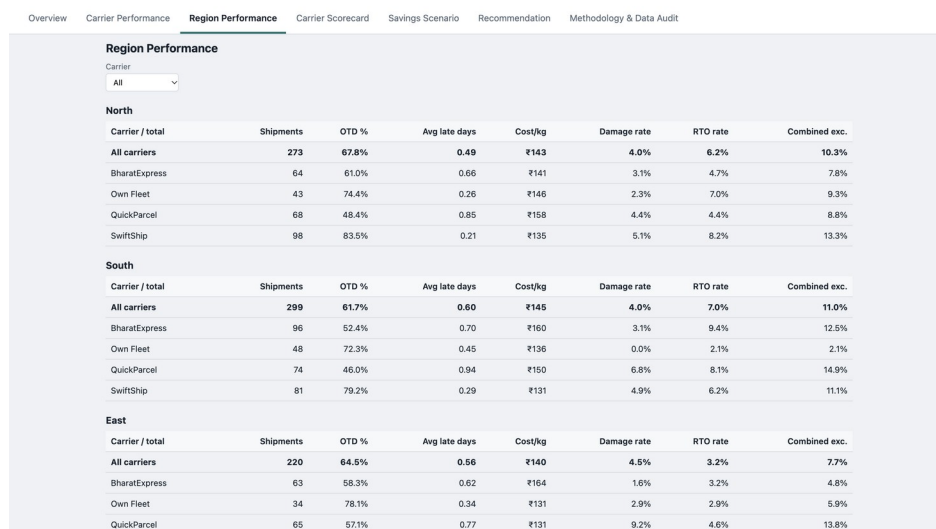


Figure 3 — Region Performance tab: all-carrier row and each carrier within North, South, East and West

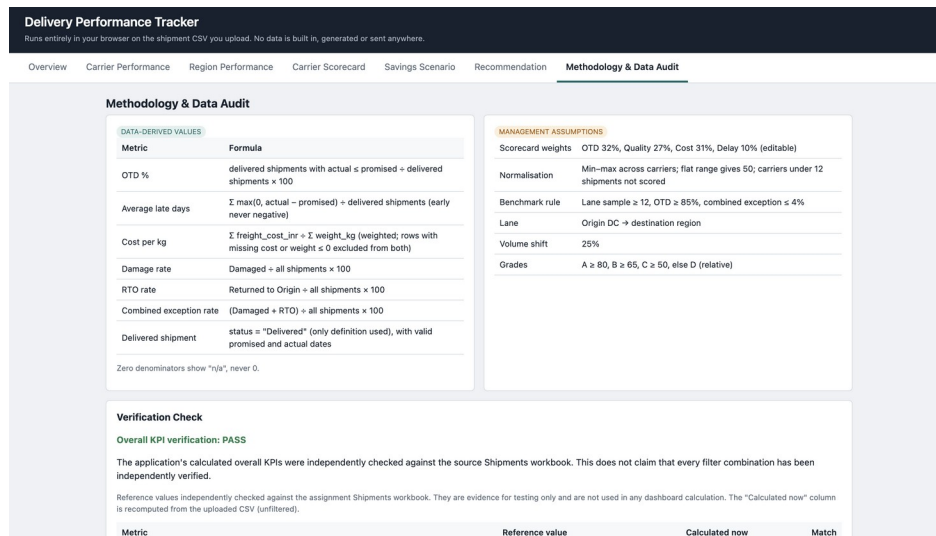


Figure 4 — Methodology & Data Audit tab: data-derived formulas, management assumptions and the Verification Check (overall KPI verification: PASS)

Note: this screenshot shows non-default settings (weights OTD 32 / Quality 27 / Cost 31 / Delay 10; minimum sample 12; 25% shift). Report tables use the defaults 30/30/30/10, minimum 10 and 50%.

1. Introduction

This report documents mini project **T12 — Delivery Performance Tracker** for the course AI Powered Tools. The brief asks for a tracker of on-time %, average delay, cost per kg and damage/RTO rate by carrier and region; a weighted carrier scorecard; an estimate of savings from shifting volume to the best carriers; and a carrier review email. Using the provided dataset of 1,200 shipments, a browser-based application was built with AI assistance (Gemini, then Claude) through three versions, and its outputs were checked against the source workbook. The brief does not name a manager, so the reader is assumed to be the **Head of Logistics / Carrier Management**.

1.1 Objectives

- Build a tracker that calculates every figure from the uploaded Shipments data, with a transparent scorecard and a savings estimate (not a guarantee).
- Verify the numbers, document AI prompts and iterations honestly, and give a specific, data-backed recommendation.

2. Dataset Overview and Data Quality

The data is synthetic and was created for teaching; it is treated as real business data. Only the Shipments sheet is used by the application. The “Gold Labels” sheet mentioned in the brief is not present in the workbook, so no gold-label accuracy score was measured.

Item	Value
Size	1,200 shipments × 13 columns; all 13 required columns present
Status	Delivered 1,095 · Returned to Origin 63 · Damaged 42
Carriers	SwiftShip 355 · BharatExpress 347 · QuickParcel 308 · Own Fleet 190
Network	4 origin DCs (Chennai, Delhi, Kolkata, Pune); 4 destination regions: North 273, South 299, East 220, West 408; 20 destination cities
Dispatch window	1 April 2025 to 24 February 2026
Upload validation (app)	1,200 records loaded; 0 duplicate shipment IDs; 0 invalid dates; no missing values in any column

Table 1 — Dataset profile and upload-validation result

No missing-data safeguard (Section 4.4) was triggered on this file. Only “Delivered” shipments enter OTD and delay; damaged and RTO shipments are counted in the exception rates.

3. AI-Assisted Development Process

Table 2 shows a planning step and four build rounds. Gemini and Claude produced the application; ChatGPT was used as a planning and logging aid (log entries 1–11). The AI Use Log holds **16 entries** (the template example row is not counted), including one rejected output and two partly kept outputs.

Round (log #)	Problem found	Prompt / process change and reason	Outcome
0 · Planning (1–4, 6)	Needed to understand the brief, rubric and which tool to use	ChatGPT used to map the rubric to build, verification and log stages; confirmed the real dataset must be supplied	Kept
1 · Methodology first (12–13, Gemini)	A build-now request would have hidden the method. Gemini’s blueprint added weights, thresholds and savings criteria that the brief does not specify	Asked for analysis only (“DO NOT build”), forbade invented columns, then asked Gemini to critique its own method and label assumptions. Result: non-negative delay, normalisation safeguards, minimum sample size for savings	Partly kept
2 · Version 1 (14, 5, 7)	Code inspection showed Version 1 generated 1,200 synthetic seeded-random records instead of using the provided data	Rejected it. Restarted in Claude with a structured prompt: inspect the workbook first, use only the Shipments sheet, CSV upload, no fallback data, ten-point self-check	Rejected
3 · Version 2 (15, Claude)	Needed a build on real data with auditable calculations	Prompt fixed the KPI formulas, scorecard and savings rules and required a list of formulas, assumptions and tests. Claude reported its own checks; recommendation and email text had not yet been reviewed in a browser	Kept

Round (log #)	Problem found	Prompt / process change and reason	Outcome
4 · Version 3 (16, 9, 11)	Version 2 offered an ambiguous option to count damaged shipments as delivered; a false date-format warning appeared on ISO dates	“Do not rebuild.” One delivered definition (status = “Delivered”); added a live Verification Check panel; fixed the false warning; scorecard/savings assumption values corrected before final screenshots	Kept as final

Table 2 — Documented prompt-improvement rounds (source: AI Use Log)

AI output was treated as something to verify. The Version 1 rejection came from reading the generated source, not from the AI’s own summary. Claude’s statements that 23 automated checks passed and that no JavaScript errors occurred (log 16) are AI-reported, so they are not counted as my verification. The distinction between sources is shown below.

Element	Origin	Final decision
KPI formulas; single “Delivered” definition; Verification Check	Refined in the Gemini review (log 13); fixed in my Claude prompts (log 15–16)	Mine: early delivery never gives negative delay
Scorecard weights 30/30/30/10; savings thresholds (sample ≥ 10 , OTD $\geq 85\%$, exceptions $\leq 4\%$)	Proposed by Gemini; not in the brief	Mine: adopted and labelled “Management assumptions – configurable”
Recommendation tab text and carrier email	Generated by fixed JavaScript templates from the calculated values (not an AI call at run time)	Actions in Section 10 are my own judgement, informed by it

Table 3 — Data-derived findings, AI suggestions and human decisions

4. Application and Dashboard Methodology

4.1 Dashboard components

The application has seven tabs: **Overview** (CSV upload, upload validation, seven KPI cards, OTD-by-carrier and exception-by-region bars, carrier and region filters); **Carrier Performance** (KPI table with average parcel weight); **Region Performance** (all-carrier row plus each carrier within North, South, East, West); **Carrier Scorecard**; **Savings Scenario**; **Recommendation** (generated text and a copyable carrier review email for a selectable carrier, defaulting to the lowest scorer); and **Methodology & Data Audit** (formulas, assumptions, Verification Check, validation).

4.2 KPI definitions

KPI	Definition as implemented
Delivered shipment	status = “Delivered” with valid promised and actual dates (the only definition used)
Operational OTD %	$\text{Delivered shipments with actual} \leq \text{promised} \div \text{delivered shipments} \times 100$
Average late days	$\Sigma \max(0, \text{actual} - \text{promised}) \div \text{delivered shipments}$
Freight cost per kg	$\Sigma \text{freight_cost_inr} \div \Sigma \text{weight_kg}$ (weighted; rows with missing cost or weight ≤ 0 excluded)
Damage / RTO / combined rate	$\text{Damaged} \div \text{all shipments}$; $\text{Returned to Origin} \div \text{all shipments}$; $(\text{Damaged} + \text{RTO}) \div \text{all shipments}$

Table 4 — KPI definitions (data-derived values)

4.3 Scorecard and savings scenario

Scorecard. Each carrier with at least 10 shipments receives four 0–100 sub-scores using min–max normalisation across carriers (higher is better for OTD; lower is better for exception rate, cost/kg and late days, so these are inverted). If all carriers have the same value, the application assigns a neutral 50. Weights are rescaled to 100% if edited; grades are relative (A ≥ 80 , B ≥ 65 , C ≥ 50 , else D). **Savings.** A lane is Origin DC → Destination Region. In each lane the benchmark is the lowest cost/kg carrier with sample ≥ 10 , OTD $\geq 85\%$ and combined exception $\leq 4\%$. For carriers costlier per kg than the benchmark, savings = cost of the shifted share – (shifted kg $\times$ benchmark cost/kg), at a selected 25%, 50% or 100% shift. Thresholds, weights and shift are management assumptions, not dataset facts.

4.4 Error handling and transparency

The application reports missing required columns and loads nothing; counts blank cells, unparseable dates and numbers, unknown status values, delivered shipments without valid dates, and duplicate IDs (reported, not removed); excludes weight ≤ 0 or missing cost rows from cost/kg; shows “n/a” rather than 0 for zero denominators; shows messages for empty filters, failed uploads and lanes with no qualifying benchmark; and never substitutes made-up values. Formulas and assumptions appear in separate “data-derived” and “management assumptions” panels.

5. Overall Analytical Results

Metric	Value	Basis
Total shipments	1,200	All records
Delivered shipments	1,095	status = "Delivered"
Operational OTD	62.37%	683 of 1,095 delivered shipments on time
Average late days	0.563	Per delivered shipment, early never negative
Freight cost per kg	₹147.56	₹1,400,559 ÷ 9,491.5 kg
Damage rate	3.50%	42 of 1,200
RTO rate	5.25%	63 of 1,200
Combined Damage/RTO rate	8.75%	105 of 1,200

Table 5 — Overall verified KPIs (cards show one decimal, e.g. 62.4%; the Verification Check shows the full values above)

Roughly four in ten delivered shipments arrive late, but average lateness is small (0.56 days): the issue is how often, not how long. About one shipment in eleven (8.75%) ends as damaged or returned.

6. Regional Performance

Region	Shipments	OTD %	Avg late days	Cost/kg	Damage %	RTO %	Combined %
North	273	67.8	0.49	₹143	4.0	6.2	10.3
South	299	61.7	0.60	₹145	4.0	7.0	11.0
East	220	64.5	0.56	₹140	4.5	3.2	7.7
West	408	58.3	0.59	₹157	2.2	4.4	6.6

Table 6 — Region performance (green = best, red = worst on that metric)

West is the largest region (408 shipments, 34%) and has both the lowest OTD (58.3%) and the highest cost, ₹157/kg, about ₹17 more per kg than East; its exception rate is the lowest (6.6%), so its problem is speed and cost, not quality. **South** has the highest exception rate (11.0%), driven by RTO (7.0%). **North** has the best OTD (67.8%) but the second-highest exceptions (10.3%). Cost/kg is not adjusted for distance or parcel weight, so cross-region cost gaps are indicative only.

7. Carrier-Level Insights

Carrier	Shipments	OTD %	Avg late days	Cost/kg	Combined exc. %	Scorecard
SwiftShip	355	76.0	0.31	₹139	9.6	79.3 (B)
Own Fleet	190	74.2	0.35	₹144	6.3	87.9 (A)
BharatExpress	347	52.8	0.70	₹154	7.2	30.0 (D)
QuickParcel	308	50.0	0.83	₹153	11.0	2.2 (D)

Table 7 — Carrier performance (network level)

Two groups emerge: SwiftShip and Own Fleet deliver on time about three times in four, while BharatExpress and QuickParcel manage about half. SwiftShip is the cheapest per kg, but its combined exception rate (9.6%) is above the network's 8.75%. QuickParcel is last on OTD, delay and exceptions. The carrier × region view shows where the gaps are sharpest:

Carrier · Region	Shipments	OTD %	Cost/kg	Combined %	Observation
Own Fleet · South	48	72.3	₹136	2.1	Lowest exception rate of all 16 cells
QuickParcel · South	74	46.0	₹150	14.9	Same region, 26 pts lower OTD
SwiftShip · North	98	83.5	₹135	13.3	Fast and cheap; RTO 8.2% blocks benchmark
QuickParcel · West	101	49.5	₹169	7.9	Highest cost/kg of all 16 cells

Table 8 — Selected carrier × region comparisons (from the Region Performance tab)

8. Carrier Scorecard

Management assumptions – configurable: OTD 30%, Quality/Exceptions 30%, Cost/kg 30%, Delay Severity 10%. These weights were proposed by AI and adopted by me; they are **not** derived statistically from the data. All four carriers have at least 10 shipments, so all are scored.

Carrier	OTD score	Quality score	Cost score	Delay score	Weighted total	Grade
Own Fleet	93	100	69	92	87.9	A
SwiftShip	100	31	100	100	79.3	B
BharatExpress	11	81	0	24	30.0	D
QuickParcel	0	0	7	0	2.2	D

Table 9 — Carrier scorecard (relative 0–100 sub-scores; Own Fleet ranks first, SwiftShip second)

Own Fleet leads because it is strong on quality (6.3% exceptions) and reliable on time; SwiftShip loses on quality. Min–max scores are relative and stretch small gaps: QuickParcel and BharatExpress differ by only 2.8 points of OTD, yet score 0 and 11 on that component. The scorecard shows an ordering under these weights, not an objectively “best” carrier, so the raw values in Table 7 should be read alongside it.

9. Savings Scenario

At the tested **50% volume shift**, the application gives **estimated/potential savings of ₹3,549**. This is a scenario estimate, not guaranteed savings (₹1,774 at 25% and ₹7,098 at 100% under the same rules). It rests on a single lane: **Delhi DC → East**, where SwiftShip qualifies as benchmark (15 shipments, OTD 86.7%, exceptions 0.0%, ₹149.4/kg) and the shifted volume comes from BharatExpress (₹200.4/kg). Eligible volume is 69.6 kg; current cost ₹13,950 would fall to ₹10,401 at benchmark cost.

- Only **1 of 16** origin-DC → region lanes has a qualifying benchmark; with “destination region only” as the lane, none qualifies and savings are ₹0.
- ₹3,549 is about **0.3% of total freight spend** (₹1,400,559), so rate shifting alone is a small lever.
- The estimate assumes unchanged cost/kg after the shift and ignores parcel mix, distance, capacity and service differences, and the benchmark rests on 15 shipments.

10. Business Recommendation

The recommendations below are for the Head of Logistics. The application’s generated text suggests reviewing QuickParcel on OTD, asking for a damage/RTO root-cause plan, and piloting a partial shift on qualifying lanes only; the prioritisation and the regional actions are my own judgement based on the figures in Sections 6–9.

#	Action	Evidence (from the tracker)	Metric to track
1	South: pilot moving part of QuickParcel’s South volume to Own Fleet, after a capacity check	South has the highest exceptions (11.0%). QuickParcel South: OTD 46.0%, 14.9% exceptions, ₹150/kg. Own Fleet South: OTD 72.3%, 2.1% exceptions, ₹136/kg. Illustration (manual, outside the app’s benchmark rule): a 50% shift of QuickParcel’s 562 kg ≈ ₹3,900 estimated/potential; it would add about 37 shipments (+77%) to Own Fleet’s 48	South OTD, exceptions and cost/kg by carrier each month
2	QuickParcel: formal performance review with a 60-day corrective plan; no new volume until OTD improves	OTD 50.0% vs network 62.4%; 0.83 late days vs 0.56; combined exceptions 11.0% vs 8.75%; scorecard 2.2/100	OTD to at least the network 62.4% (proposed target)
3	West: review BharatExpress and QuickParcel; use the review to renegotiate rates	West is 34% of shipments with the lowest OTD (58.3%) and ₹157/kg. BharatExpress West OTD 46.2% (124 shipments); QuickParcel West ₹169/kg (101)	West OTD and cost/kg
4	North / SwiftShip: ask for an RTO root-cause plan (address checks, delivery attempts) before giving more volume	SwiftShip North: OTD 83.5%, ₹135/kg, but 13.3% exceptions (RTO 8.2%), so it misses the ≤ 4% benchmark	North SwiftShip exceptions toward ≤ 4%
5	Savings: treat ₹3,549 as a small pilot (Delhi DC → East); do not move all volume to one carrier	One qualifying lane; 0.3% of spend; benchmark sample of 15	Actual cost/kg and OTD in that lane after the pilot

Table 10 — Prioritised recommendations. Targets are proposals; all savings are estimated/potential, not guaranteed

Overall, service reliability is a larger lever than rate: no carrier reaches 85% OTD network-wide (best: SwiftShip 76.0%), so fixing QuickParcel and BharatExpress matters more than reallocating volume.

11. Verification and Correctness

Check	What was checked	Result
Overall KPI verification (app)	8 values (shipments, delivered, OTD, late days, cost/kg, damage, RTO, combined) recomputed from the uploaded CSV and compared with reference values independently checked against the Shipments workbook	PASS – 8/8 matched
Upload validation (app)	Records, 13 required columns, duplicate IDs, invalid dates, missing values	1,200 · 13/13 · 0 · 0 · none
Savings scenario	Savings recalculated independently at the 50% shift	₹3,549 reproduced
Regional, carrier, scorecard figures	Spot-checked against the source data (selected values, not every filter combination)	Matched where checked
AI-reported tests (log 15–16)	Claude’s browser smoke test and 23 automated checks	Reported by the AI; not independent

Table 11 — Verification evidence and its limits

The reference values in the Verification Check are stored in the application for testing only and are never used in its calculations. Because they are the same values supplied to Claude in the Version 3 prompt, PASS shows the final build still reproduces the earlier verified numbers; it is a consistency check, not a new independent audit. **Verification was not exhaustive:** every carrier × region × filter combination was not independently checked, and with no Gold Labels sheet no gold-label accuracy was measured. While preparing this report, the regional, carrier, scorecard and savings figures in Sections 6–9 were also recomputed from the Shipments sheet and matched the application.

12. Responsible AI and Limitations

- **Errors found and corrected:** Version 1 used synthetic data (rejected); AI-proposed weights and thresholds were not in the brief (relabelled as assumptions); Version 2’s ambiguous delivered option was removed; a false date-format warning was fixed.
- **AI limitations:** models assumed columns, generated synthetic data, and reported their own checks; each needed inspection.
- **Management assumptions:** scorecard weights, grades, benchmark thresholds, lane definition and shift percentage are configurable choices; changing them changes rankings and savings.
- **Savings and data limits:** a one-lane estimate, no adjustment for parcel mix, distance or capacity, small cells (10–15 shipments), and 1,200 synthetic records over under a year.
- **Verification limits:** see Section 11. No personal or confidential data was used; the dataset is synthetic.

13. Personal Reflection

The biggest lesson was that a polished answer is not a correct one. Version 1 looked complete, but only reading the code showed it had generated 1,200 random shipments instead of using mine. That changed how I prompted: the Gemini prompt that said “do not build, first analyse” gave me a method I could question, and the Claude prompt that named the data rules, forbade fallback data and asked for a self-check produced something I could test.

I also learned that AI will happily fill gaps with assumptions. The 30/30/30/10 weights and the 85% / 4% thresholds came from the AI, so I labelled them as management assumptions instead of facts. The savings result humbled me: ₹3,549 from one lane and about 0.3% of spend. It showed that the real insight was about service quality, which the tracker only made visible after I compared carriers within regions. Judging what to recommend, and how cautiously to word it, was my job, not the tool’s. Next time I would log each prompt as I go, test smaller filter cases earlier, and treat the AI’s own test reports as claims to check.

14. Contributions and AI Disclosure

Contributor / tool	Contribution
Student	Prompts, rejection of Version 1, decisions, verification, recommendations, AI Use Log, review of report
Gemini Canvas (log 12–14)	Method blueprint and critique; Version 1 (rejected)
Claude (log 15–16)	Versions 2 and 3 of the tracker (final HTML)
ChatGPT (log 1–11)	Assignment planning, prompt drafting, log-entry structuring
Claude (report)	Drafting this report from the HTML, workbook and log; recomputing figures for cross-checking

Table 12 — Contributors and AI disclosure

Appendix A — Supporting Screenshots

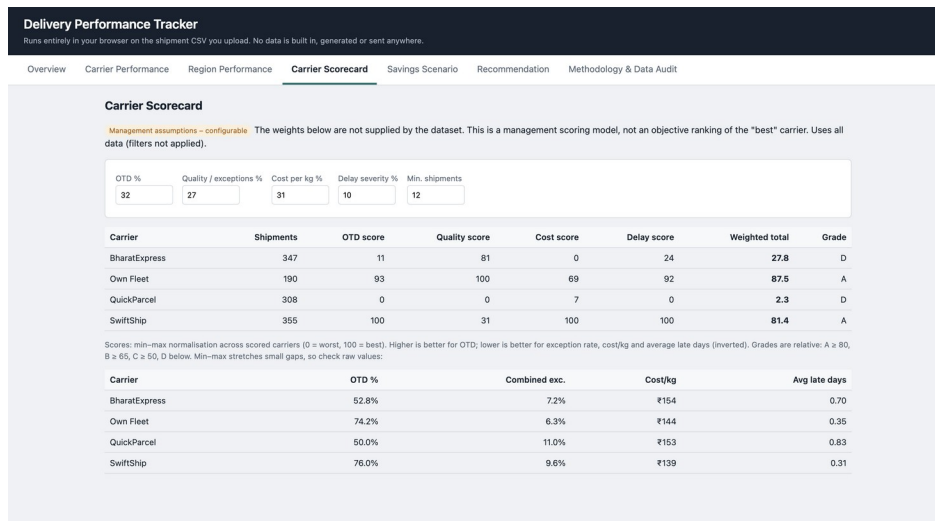


Figure 5 — Carrier Scorecard tab with the “Management assumptions – configurable” label, editable weights and raw-value table

Note: this screenshot shows non-default settings (weights 32 / 27 / 31 / 10; minimum shipments 12). Report Table 9 uses the defaults 30/30/30/10 and minimum 10.

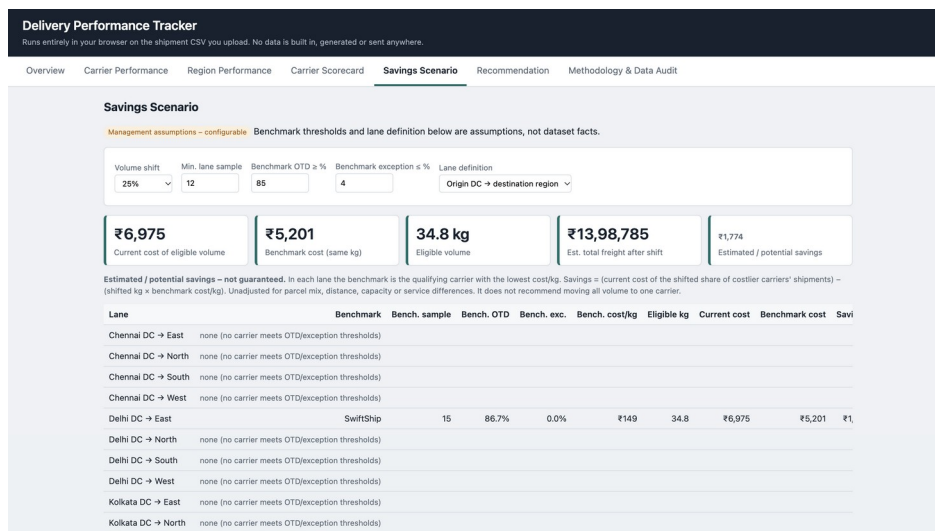


Figure 6 — Savings Scenario tab: configurable benchmark rules and estimated / potential savings (not guaranteed); one qualifying lane, Delhi DC → East

Note: this screenshot shows non-default settings (25% shift; minimum lane sample 12). Report Section 9 quotes the 50% shift (₹3,549) with minimum sample 10; this view shows ₹1,774.

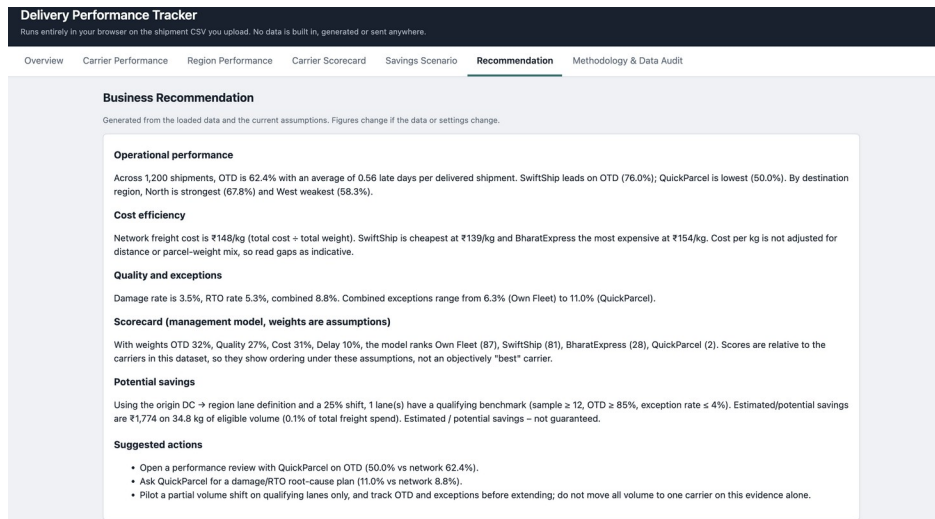


Figure 7 — Recommendation tab: text generated from the calculated values

Note: this screenshot shows non-default settings (weights 32 / 27 / 31 / 10; minimum 12; 25% shift).

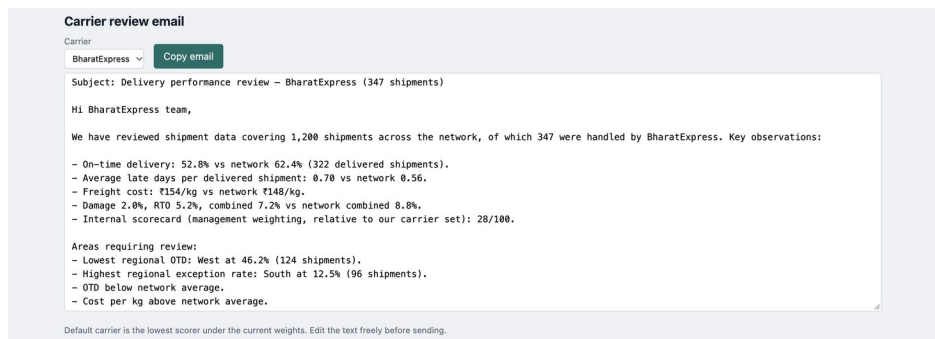


Figure 8 — Carrier review email generated by the application (BharatExpress selected)

Note: this screenshot shows non-default settings (weights 32 / 27 / 31 / 10, which give BharatExpress a scorecard of 28/100).